

Deep Learning-Based Detection of Oral Cancer from RGB Clinical Images

*Thesis Submitted for Partial Fulfillment of the
Requirements for the Award of the Degree of*

Bachelor of Technology in Electronics and
Communication Engineering

by

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21st April, 2026

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This is to certify that the thesis entitled “Deep Learning-Based Detection of Oral Cancer from RGB Clinical Images” submitted by Gautam Giri to Indian Institute of Information Technology, Guwahati, is a record of bonafide research work under my supervision, and I consider it worthy of consideration for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering at Indian Institute of Information Technology Guwahati.

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Declaration

I declare that

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2. The work has not been submitted to any other Institute for any degree or diploma.
3. I have followed the guidelines provided by the Institute in writing the thesis.
4. I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
5. Whenever I have used materials (data, theoretical analysis, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references.
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Gautam Giri

Abstract

Artificial Intelligence (AI) has emerged as a powerful tool in medical image analysis, enabling early detection and diagnosis of various diseases. This project focuses on the development of a deep learning-based system for the classification of oral lesions using clinical images. Early detection of oral cancer is crucial, as delayed diagnosis significantly reduces survival rates and treatment effectiveness.

In this work, multiple convolutional neural network (CNN) architectures are explored, including LeNet, VGG16, and MobileNetV3. LeNet is used as a baseline model for initial experimentation, while transfer learning-based architectures such as VGG16 and MobileNetV3 are utilized to extract high-level features from intraoral images. The models are trained and validated on a dataset consisting of normal and cancerous oral images, with preprocessing techniques applied to improve image quality and consistency.

To enhance model reliability and generalization, K-Fold cross-validation is implemented, reducing the chances of overfitting and ensuring robust evaluation. The performance of the models is assessed using metrics such as accuracy and validation scores, demonstrating the effectiveness of deep learning in distinguishing between different oral conditions. Previous studies have shown that deep learning models can achieve performance comparable to medical experts in detecting oral cancer from photographic images.

In addition to model development, a web-based application is designed for real-time deployment. The system allows users to upload oral images through an intuitive interface, and the trained model provides instant predictions. This demonstrates the practical applicability of the solution in real-world scenarios.

The proposed system contributes toward the development of AI-assisted diagnostic tools that can support early screening and improve healthcare accessibility, particularly in resource-limited environments. Future work includes expanding the dataset, improving model performance, and deploying the system on mobile platforms for wider adoption.

Chapter 1: Introduction

Oral cancer is one of the most common types of cancer worldwide and poses a significant public health challenge. Early detection plays a crucial role in improving survival rates, as late-stage diagnosis often leads to poor prognosis and limited treatment options. However, identifying oral lesions at an early stage is difficult due to their diverse clinical appearance and dependence on expert evaluation.

With the advancement of Artificial Intelligence (AI), particularly in the field of deep learning, automated analysis of medical images has gained significant attention. Convolutional Neural Networks (CNNs) have proven to be highly effective in extracting complex patterns from images and have been successfully applied in various medical domains, including cancer detection. In the context of oral lesion classification, AI-based systems can assist healthcare professionals by providing faster and more consistent diagnoses.

In this project, a deep learning-based approach is proposed for the classification of oral cavity images into normal and cancerous categories. Multiple CNN architectures are explored, including LeNet as a baseline model and advanced transfer learning models such as VGG16 and MobileNetV3. These models are trained on a dataset of intra-oral images, enabling the system to learn distinguishing features of different oral conditions.



Figure 1: Overview of the proposed AI-based oral lesion classification system

To ensure robustness and improve generalization, K-Fold cross-validation is employed during training. This technique helps in better utilization of the available dataset and reduces the risk of overfitting. The performance of the models is evaluated using standard metrics such as accuracy and validation scores.

Furthermore, to demonstrate the practical applicability of the system, a web-based deployment is implemented. The application allows users to upload images and receive predictions in real time, making the system accessible and user-friendly. Such AI-driven solutions have the potential to support early screening and assist in clinical decision-making.

The remainder of this report is organized as follows: Chapter 2 discusses the motivation behind the work, Chapter 3 outlines the objectives, Chapter 4 describes the methodology and implementation, and Chapter 5 presents the conclusion and future scope.

Chapter 2: Motivation

The early detection of oral cancer remains a critical challenge in healthcare, primarily due to the lack of awareness, limited access to specialists, and the subtle visual differences between normal and abnormal oral lesions. In many cases, oral cancer is diagnosed at advanced stages, significantly reducing the chances of successful treatment and survival. This highlights the urgent need for automated and accessible diagnostic support systems.

Recent advancements in Artificial Intelligence (AI), particularly deep learning, have shown promising results in medical image classification. Convolutional Neural Networks (CNNs) have demonstrated the ability to learn complex visual patterns and assist in disease detection with high accuracy. These capabilities motivate the development of an AI-based system that can analyze oral cavity images and support early screening.

Another key motivation for this work is the increasing availability of smartphone cameras and web-based technologies. Integrating AI models with web applications can enable real-time diagnosis and make such systems accessible even in resource-limited settings, where access to expert medical professionals is limited.

Additionally, this project provides an opportunity to explore and compare different CNN architectures such as LeNet, VGG16, and MobileNetV3. While LeNet serves as a simple baseline model, transfer learning-based architectures can leverage pre-trained knowledge to improve performance on limited datasets. This comparative approach helps in understanding the trade-offs between model complexity, accuracy, and computational efficiency.

From the analysis of existing research works, the following key gaps relevant to this project have been identified:

- **Limited Dataset Availability**
Most studies use small or controlled datasets, limiting real-world generalization .
- **Dependence on Image Quality**
Models are often trained on high-quality images, making them less reliable on real-world data.
- **Lack of Practical Deployment**
Few works implement real-time systems such as web or mobile applications.
- **Limited Use of Lightweight Models**
Many models are computationally heavy, restricting deployment on low-resource devices.

Chapter 3: Objective

The objective of this work is to design and develop an efficient and practical system for the classification of oral lesion images using deep learning techniques. This study focuses not only on achieving accurate predictions but also on addressing real-world challenges such as limited data, varying image conditions, and deployment feasibility. The specific objectives of this work are as follows:

- To develop a deep learning-based classification system:
The system is designed to classify oral lesion images into normal and cancerous categories using convolutional neural networks. Multiple architectures such as LeNet, VGG16, and MobileNetV3 are explored to understand their effectiveness.
- To address the challenge of limited dataset availability:
Medical datasets are often small and imbalanced. To overcome this, preprocessing and data augmentation techniques are applied, along with K-Fold cross-validation to improve model generalization and reduce overfitting.
- To improve robustness for real-world image conditions:
The model is trained on images with variations in lighting, resolution, and orientation, ensuring reliable performance in practical, non-ideal scenarios.
- To implement efficient and lightweight models:
Lightweight architectures such as MobileNetV3 are used to reduce computational complexity, making the system suitable for deployment on low-resource devices.
- To enable practical deployment through a web-based application:
A web interface is developed to allow users to upload images and obtain predictions in real time, demonstrating real-world applicability.
- To evaluate and compare different CNN architectures:
The performance of different models is analyzed using evaluation metrics such as accuracy and validation scores to identify the most suitable approach.

Chapter 4 Work Done

This chapter describes the methodology and implementation of the proposed system for oral lesion classification. The work involves designing a deep learning pipeline, training multiple convolutional neural network models, and deploying the final model through a web-based application. The overall approach focuses on achieving accurate classification while ensuring robustness and practical usability.

4.1 Proposed Technique

The proposed technique is based on the use of convolutional neural networks (CNNs) for automatic classification of oral lesion images. The system takes an input image of the oral cavity and processes it through a trained deep learning model to predict whether the image belongs to a normal or cancerous category.

The workflow of the proposed system consists of the following steps:

- **Image Acquisition:**
The dataset consists of intra-oral images categorized into normal and cancer classes.
- **Preprocessing:**
The images are resized and normalized to match the input requirements of the neural networks. This ensures consistency across the dataset.
- **Data Augmentation:**
Techniques such as rotation, flipping, and scaling are applied to increase dataset diversity and reduce overfitting.
- **Model Selection:**
Multiple CNN architectures are used:
 - LeNet as a baseline model
 - VGG16 for deep feature extraction
 - MobileNetV3 for efficient and lightweight performance
- **Training Strategy:**
K-Fold cross-validation is applied to improve generalization and evaluate model performance on different data splits.
- **Prediction:**
The trained model outputs the predicted class of the input image.

This approach is inspired by existing AI-based classification systems that utilize CNNs for medical image analysis and have demonstrated high accuracy in detecting oral lesions

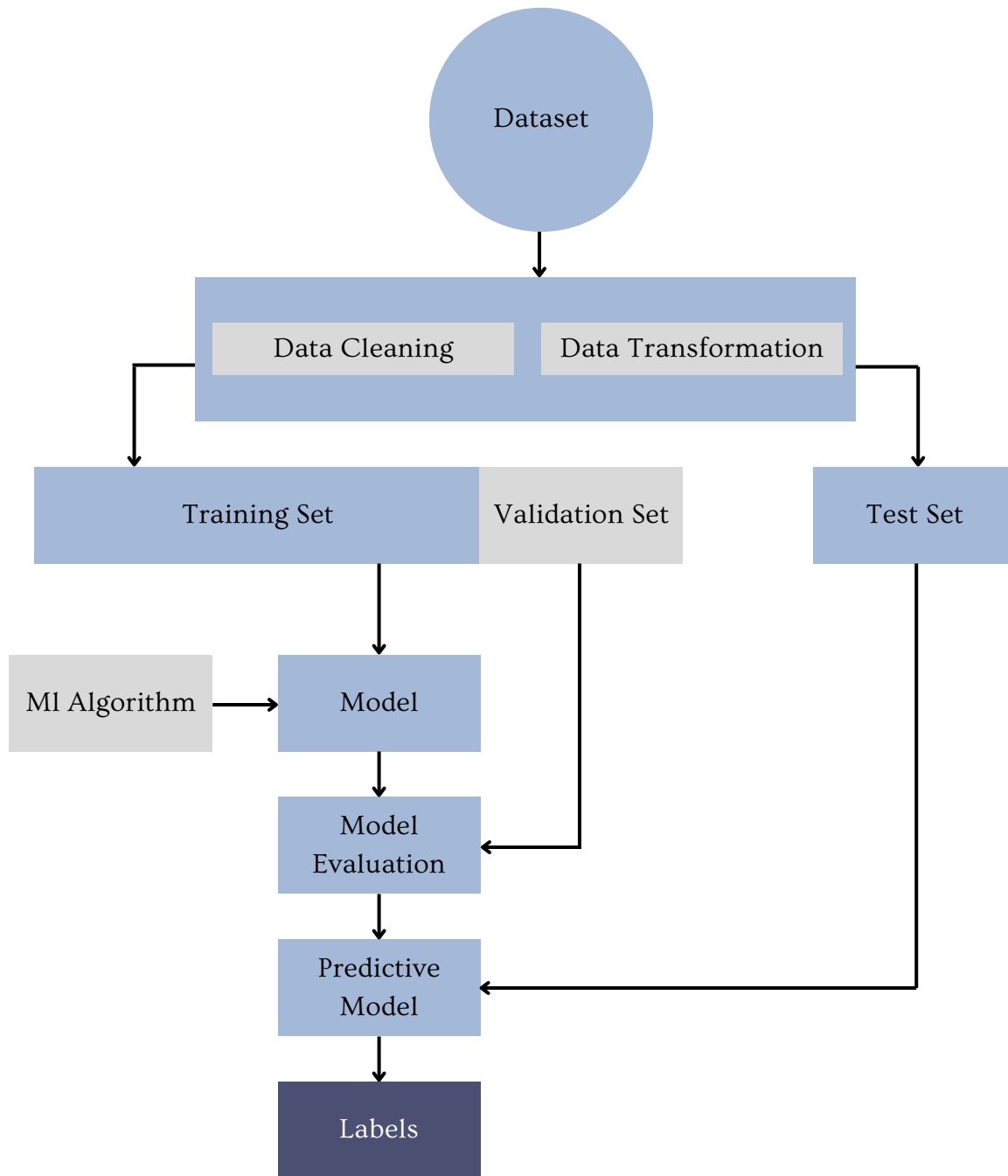


Figure 4.1: Workflow of the proposed oral lesion classification system

4.2 Implementation of Proposed Technique

The implementation of the proposed system involves dataset preparation, model training, evaluation, and deployment.

4.2.1 Dataset Preparation

The dataset consists of oral cavity images belonging to different classes. Since the dataset size is limited, preprocessing and augmentation techniques are applied to improve performance. Images are resized to a fixed dimension and normalized before being fed into the models.

To further enhance dataset diversity and reduce overfitting, several data augmentation techniques are applied. These include rotation, where images are rotated by small angles to simulate different orientations; horizontal flipping, which helps the model generalize across symmetric variations; and scaling/zooming, which allows the model to learn features at different magnifications. Additionally, minor variations in brightness and contrast are introduced to account for different lighting conditions during image capture.

These augmentation techniques help the model become more robust to real-world variations in image quality, thereby improving generalization and overall performance.

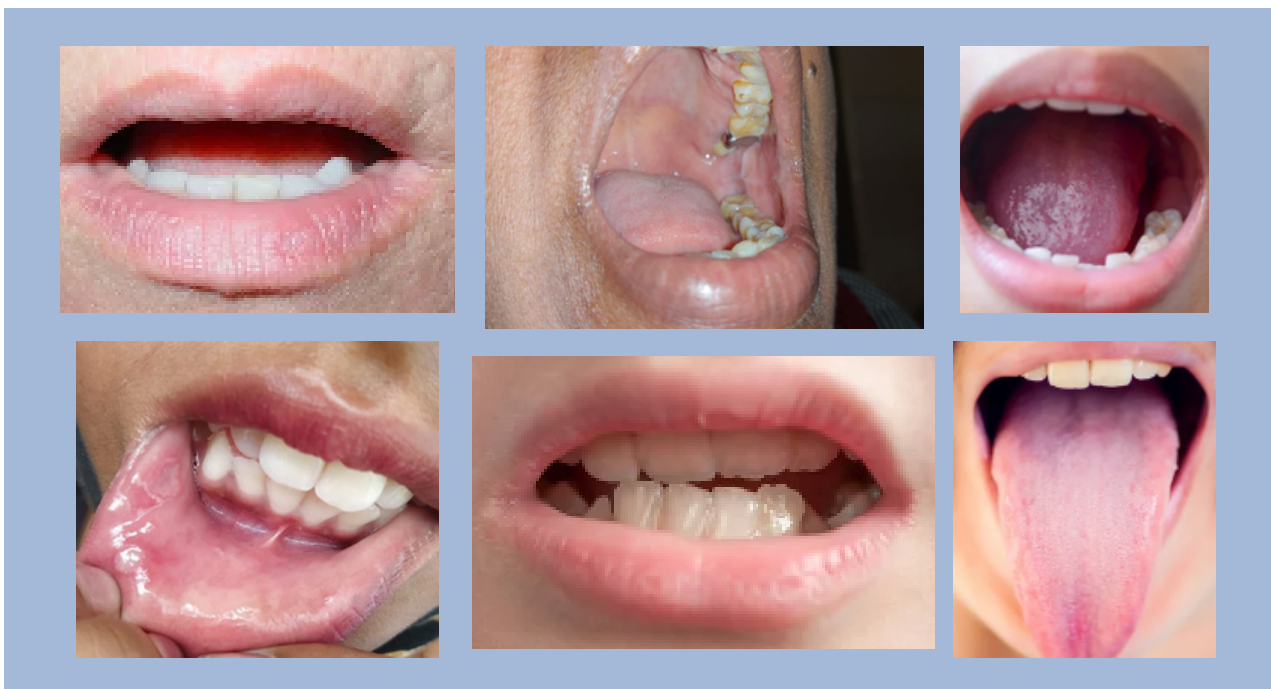


Figure 4.2: various images (both cancerous and normal) used in training of models.

4.2.2 Model Development

Three CNN architectures are implemented:

- LeNet: LeNet consists of alternating convolution and pooling layers followed by fully connected layers. The input image (128×128) is passed through convolution layers for feature extraction and pooling layers for dimensionality reduction. The output is flattened and passed through fully connected layers, where an additional custom layer is introduced before final classification.

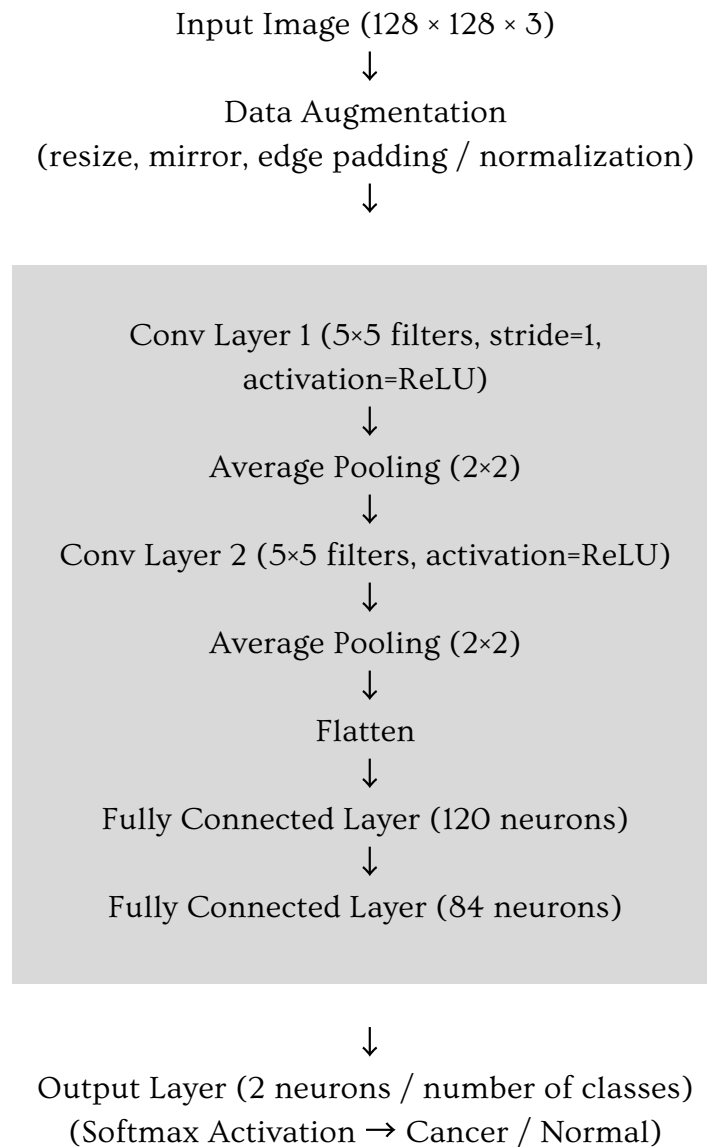


Figure 4.3: LeNet architecture for oral image classification.

- MobileNetV3: MobileNetV3 uses depthwise separable convolutions and inverted residual bottleneck blocks to reduce computational complexity. It also includes squeeze-and-excitation modules for better feature representation. Transfer learning is applied with initial layer freezing followed by partial fine-tuning.



Figure 4.4: MobileNetV3 architecture with transfer learning and fine-tuning.

- **VGG16:** VGG16 is a deep CNN composed of multiple convolutional blocks with 3×3 filters, followed by max-pooling layers. The extracted features are flattened and passed through fully connected layers. In this work, transfer learning is applied by freezing initial layers and fine-tuning later layers for classification.

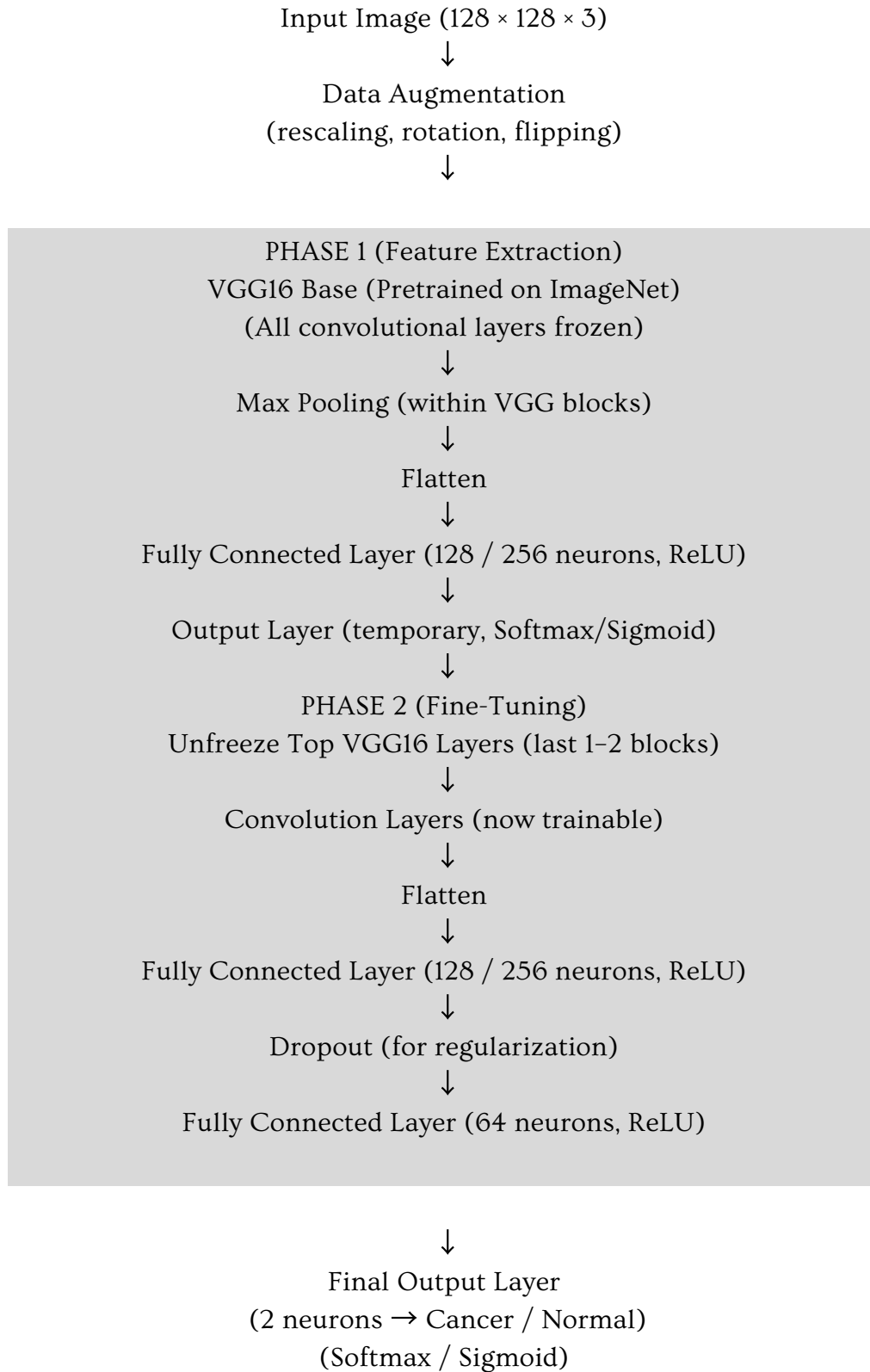


Figure 4.5: VGG16-based transfer learning architecture with fine-tuning.

4.2.3 Training and Validation

The models are trained using K-Fold cross-validation to ensure robustness. The dataset is divided into multiple folds, and the model is trained and validated on different subsets to reduce bias.

Evaluation metrics include:

- Accuracy
- Validation accuracy
- Loss

This approach helps in selecting the best-performing model.

4.2.4 Results and Analysis

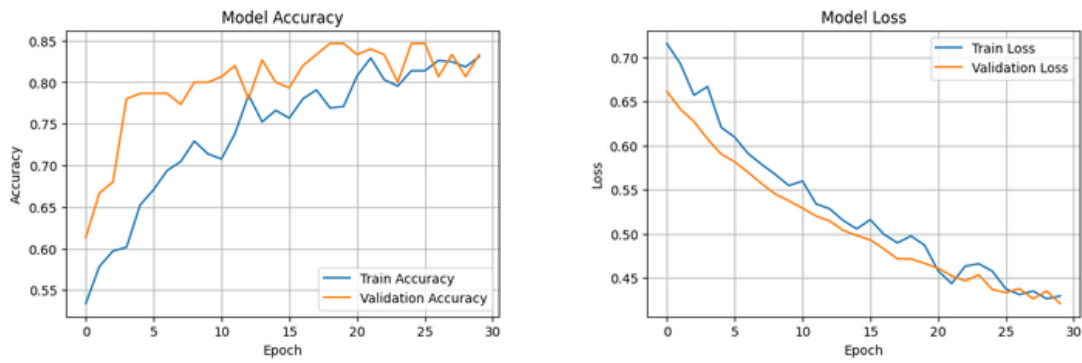


Figure 4.6: Training and validation curves for the VGG16 model with all layers frozen,

Figure 4.6 shows the training and validation accuracy and loss curves for VGG16 with all layers frozen. Accuracy increases and loss decreases steadily, indicating stable learning with minimal overfitting. This aligns with Table 4.1, where VGG16 achieves 83.68% validation accuracy with low loss.

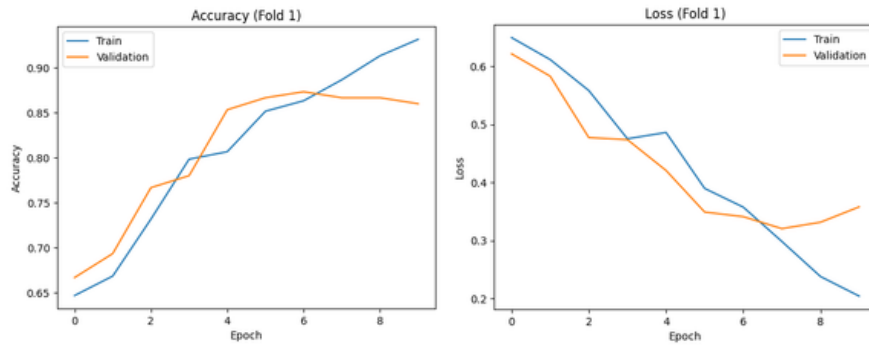


Figure 4.7: Training and validation loss curves for the best-performing fold of the LeNet model.

Figure 4.7 shows the training and validation loss curves for the best-performing fold of the LeNet model. The decreasing loss indicates effective learning, while slight variations suggest lower generalization. This is reflected in Table 4.1, where LeNet achieves 78.67% validation accuracy with higher loss than VGG16.

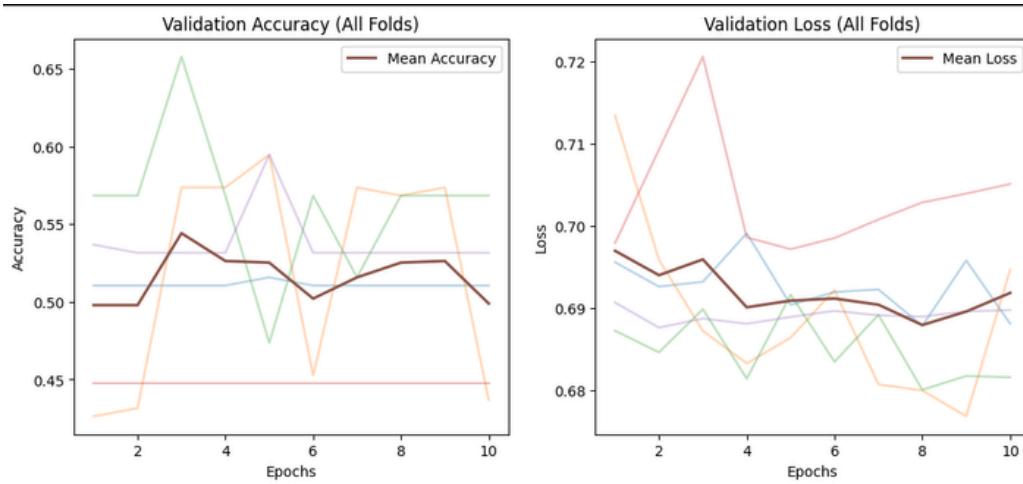


Figure 4.8: validation accuracy and loss curves for the MobileNetV3 model, indicating effective learning and steady convergence over epochs.

Figure 4.8 illustrates the validation accuracy and loss across all folds for the MobileNetV3 model. The model shows moderate improvement in accuracy and gradual reduction in loss, but the fluctuations across folds indicate variability in performance, likely due to limited dataset size. As shown in Table 4.1, MobileNetV3 achieves a validation accuracy of 53.16%, which is significantly lower than the other models, along with comparatively higher loss.

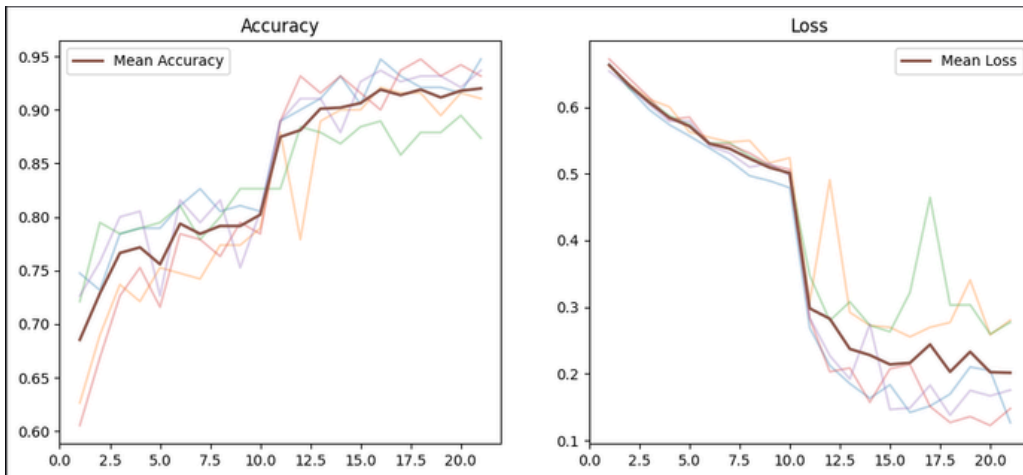


Figure 4.9: Validation accuracy and loss across all folds for the VGG16 model using transfer learning, showing stable performance and consistent convergence.

Figure 4.9 presents the validation accuracy and loss across all folds for the VGG16 model using transfer learning. The model demonstrates consistent performance with higher accuracy and lower loss compared to other architectures. The smooth convergence across folds highlights its strong feature extraction capability and better generalization. This is further supported by Table 4.1, where VGG16 achieves the highest validation accuracy of 91.68% among all models.

Model	Training Accuracy	Validation Accuracy	Loss
VGG16	94.45%	91.68%	18.28%
LeNet	92.27%	78.67%	60.53%
MobileNetV3	72.15%	53.16%	68.68%

Table 4.1: Performance comparison of different CNN models

4.2.5 Web Deployment

To make the system practically usable, a web-based application is developed. The trained model is integrated into the backend, allowing users to upload images and receive predictions in real time.

The deployment system works as follows:

- User uploads an image
- Image is preprocessed
- Model performs inference
- Result is displayed on the interface

In addition to the predicted class, the application also displays the confidence score (probability) associated with the prediction, providing a measure of the model's certainty. This bridges the gap between theoretical model development and real-world application.

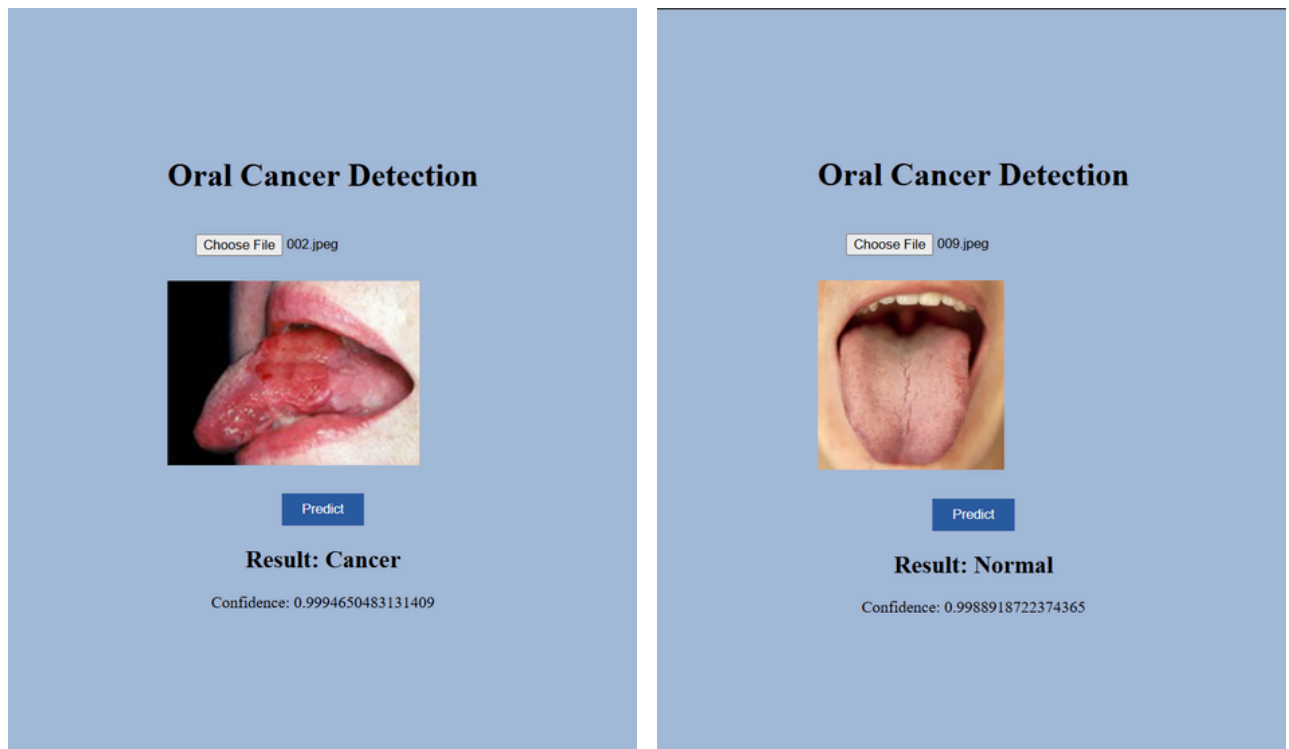


Figure 4.10: Sample web application outputs showing predicted class and confidence scores.

Chapter 5 Conclusion and Future Work

5.1 Conclusion

In this work, a deep learning-based system for the classification of oral lesion images has been successfully developed using convolutional neural network architectures such as LeNet, VGG16, and MobileNetV3. The models were trained and evaluated using preprocessing techniques, data augmentation, and K-Fold cross-validation to improve generalization and performance.

Among the implemented models, VGG16 demonstrated the best performance in terms of validation accuracy and loss, highlighting its effectiveness in extracting meaningful features from medical images. The system was also tested under varying image conditions, ensuring its robustness in real-world scenarios.

In addition to model development, a web-based application was implemented to enable real-time image classification. This demonstrates the practical applicability of the proposed system and bridges the gap between theoretical research and real-world deployment. Overall, the proposed approach shows promising potential as an assistive tool for early detection and classification of oral lesions.

5.2 Future Work

Although the proposed system achieves satisfactory performance, several improvements can be made in the future. One important direction is the use of larger and more diverse datasets to further improve model accuracy and generalization. Incorporating more classes beyond binary classification (such as different types of lesions) can enhance the clinical relevance of the system.

Another potential improvement is the integration of additional evaluation metrics such as precision, recall, and F1-score for more comprehensive analysis. The model can also be optimized further for deployment on mobile devices using techniques such as model compression or TensorFlow Lite.

Future work may also include improving the web application by enhancing the user interface and adding features such as result history and confidence scores. Additionally, combining image-based predictions with clinical data could significantly improve diagnostic reliability.

Overall, the system can be extended into a more advanced and reliable tool for assisting healthcare professionals in early diagnosis and screening of oral diseases.

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